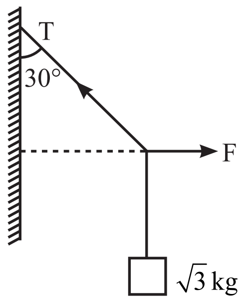
Manzil JEE - Practice (2025)

Laws of Motion

Most Important JEE PYQs

Single Correct Type Questions

1. A block of $\sqrt{3}$ kg is attached to a string whose other end is attached to the wall. An unknown force F is applied so that the string makes an angle of 30º with the wall. The tension T in the string is:
(Given $g = 10$ ms^{-2}) **[30 Jan, 2023 (Shift-II)]**



(*a*) 20 N (*b*) 25 N (*c*) 10 N (*d*) 15 N

2. At any instant the velocity of a particle of mass 500 *g* is $\left(2t\hat{i} + 3t^2\hat{j}\right) ms^{-1}$. If the force acting on the particle at $t = 1$s is $\left(\hat{i} + x\hat{j}\right)N$. Then the value of x will be: **[08 Apr, 2023 (Shift-1)]**

(*a*) 3 (*b*) 4 (*c*) 6 (*d*) 2

3. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): An electric fan continues to rotate for some time after the current is switched off.

Reason (R): Fan continuous to rotate due to inertia of motion.

In the light of above statements, choose the most appropriate answer from the options given below. **[10 Apr, 2023 (Shift-II)]**

(*a*) A is correct but R is not correct.

(*b*) Both A and R are correct and R is the correct explanation of A.

(*c*) A is not correct but R is correct.

(*d*) Both A and R are correct but R is not the correct explanation of A.

4. A particle of mass m is moving in the xy-plane such that its velocity at a point (x, y) is given as $\vec{v} = \alpha(y\hat{x} + 2x\hat{y})$, where α is a non-zero constant. What is the force $\vec{F}$ acting on the particle? **[JEE Adv, 2023]**

(*a*) $\vec{F} = 2m\alpha^2(x\hat{x} + y\hat{y})$

(*b*) $\vec{F} = m\alpha^2(y\hat{x} + 2x\hat{y})$

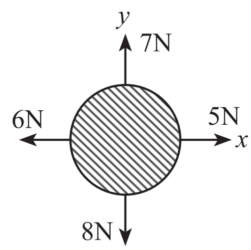
(*c*) $\vec{F} = 2m\alpha^2(y\hat{x} + x\hat{y})$

(*d*) $\vec{F} = m\alpha^2(x\hat{x} + 2y\hat{y})$

5. A mass of 10 kg is suspended by a rope of length 4m, from the ceiling. A force F is applied horizontally at the mid-point of the rope such that the top half of the rope makes an angle of 45° with the vertical. Then F equals
(Take $g = 10$ ms^{-2} and the rope to be massless)
[7 Jan, 2020 (Shift-II)]

(*a*) 75 N (*b*) 90 N (*c*) 100 N (*d*) 70 N

6. For a free body diagram shown in the figure, the four forces are applied in the 'x' and 'y' directions. What additional force must be applied and at what angle with positive x-axis so that the net acceleration of body is zero? **[25 July, 2022 (Shift-II)]**



(*a*) $\sqrt{2}$N, 45^o (*b*) $\sqrt{2}$N, 135^o

(*c*) $\frac{2}{\sqrt{3}}$N, 30^o (*d*) 2N, 45°

7. A monkey of mass 50 kg climbs on a rope which can withstand the tension (T) of 350 N. If monkey initially climbs down with an acceleration of 4 m/s^2 and then climbs up with an acceleration of 5 m/s^2. Choose the correct option
(Take $g = 10$ ms^{-2}) **[26 July, 2022 (Shift-I)]**

(*a*) $T = 700$ N while climbing upward

(*b*) $T = 350$ N while going downward

(*c*) Rope will break while climbing upward

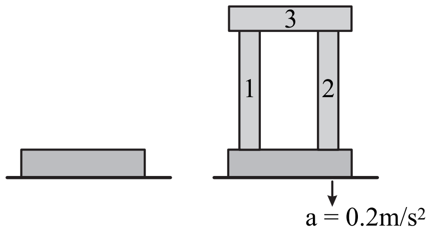
(*d*) Rope will break while going downward

8. A force $\vec{F} = \left(40\hat{i} + 10\hat{j}\right)N$ acts on a body of mass 5 kg. If the body starts from rest, its position vector $\vec{r}$ at time t =10s, will be: **[25 July, 2021 (Shift-II)]**

(*a*) $\left(100\hat{i} + 100\hat{j}\right)m$ (*b*) $\left(400\hat{i} + 100\hat{j}\right)m$

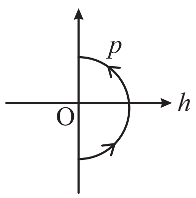
(*c*) $\left(400\hat{i} + 400\hat{j}\right)m$ (*d*) $\left(100\hat{i} + 400\hat{j}\right)m$

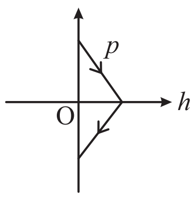
9. A steel block of 10 kg rests on a horizontal floor as shown. When three iron cylinders are placed on it as shown, the block and cylinders go down with an acceleration 0.2m/s^2. The normal reaction R' by the floor if mass of the iron cylinders are equal and of 20kg each, is _________N.
[Take g = 10m/s^2 and $\mu_s = 0.2$] **[20 July, 2021 (Shift-I)]**

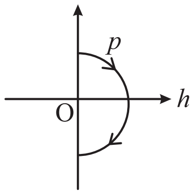


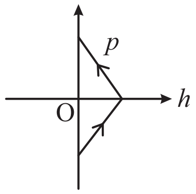
(*a*) 714 (*b*) 684 (*c*) 716 (*d*) 686

10. A ball is thrown vertically up (taken as +z-axis) from the ground. The correct momentum height (p-h) diagram is:
[9 April, 2019 (Shift-I)]

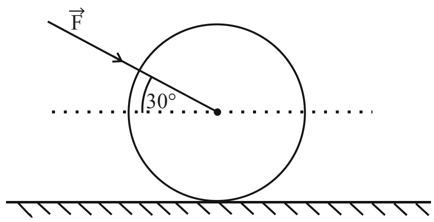
(*a*)


(*b*)


(*c*)


(*d*)


11. As shown in figure, a 70kg garden roller is pushed with a force of $\vec{F} = 200\,N$ at an angle of 30º with horizontal. The normal reaction on the roller is (Given $g = 10$ ms^{-2})

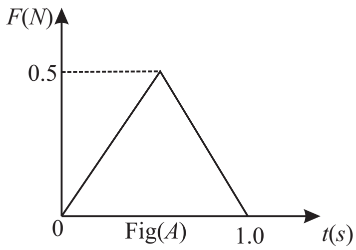


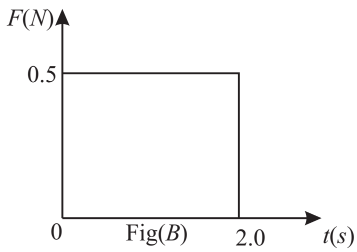
[31 Jan, 2023 (Shift-I)]

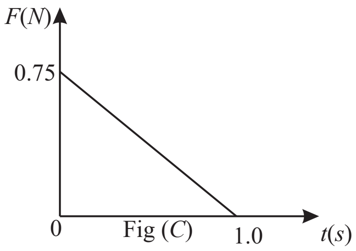
(*a*) $800\sqrt{2}\,N$ (*b*) $600\,N$

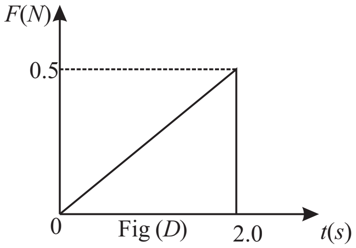
(*c*) $800\,N$ (*d*) $200\sqrt{3}\,N$

12. Figures (A), (B), (C) and (D) show variation of force with time.









The impulse is highest in figure.

[1 Feb, 2023 (Shift-II)]

(*a*) Fig (*C*)

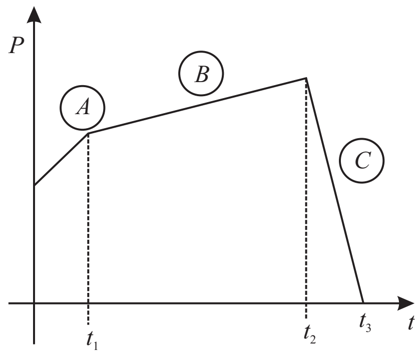
(*b*) Fig (*B*)

(*c*) Fig (*A*)

(*d*) Fig (*D*)

13. The figure represents the momentum time $(P - t)$ curve for a particle moving along an axis under the influence of the force. Identify the regions on the graph where the magnitude of the force is maximum and minimum respectively?

If $(t_3 - t_2) < t_1$. **[30 Jan, 2023 (Shift-I)]**



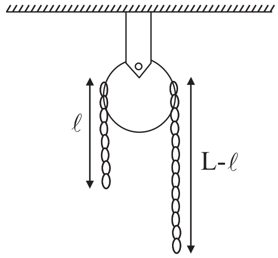
(*a*) *C* and *A* (*b*) *B* and *C*
(*c*) *C* and *B* (*d*) *A* and *B*

14. 100 balls each of mass m moving with speed v simultaneously strike a wall normally and are reflected back with the same speed, in time t s. The total force exerted by the balls on the wall is **[31 Jan, 2023 (Shift-I)]**

(*a*) $\frac{100mv}{t}$ (*b*) $\frac{200mv}{t}$ (*c*) $200\ mvt$ (*d*) $\frac{mv}{100t}$

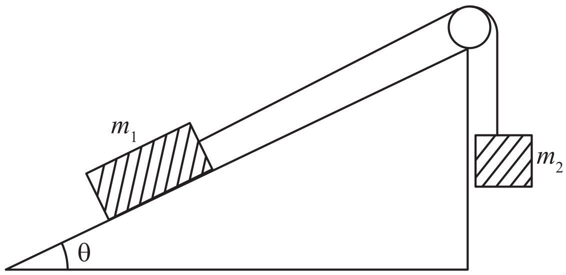
15. A uniform metal chain of mass m and length 'L' passes over a massless and frictionless pulley. It is released from rest with a part of its length 'ℓ' is hanging on one side and rest of its length '$L - \ell$' is hanging on the other side of the pully. At a certain point of time, when $\ell = \frac{L}{x}$ the acceleration of the chain is $\frac{g}{2}$. The value of x is ________.

[28 July, 2022 (Shift-II)]



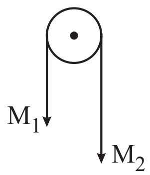
(*a*) 6
(*b*) 2
(*c*) 1.5
(*d*) 4

16. Two bodies of masses $m_1 = 5$kg and $m_2 = 3$kg are connected by a light string going over a smooth light pulley on a smooth inclined plane as shown in the figure. The system is at rest. The force exerted by the inclined plane on the body of mass m_1 will be: [Take $g = 10\ ms^{-2}$]
[29 July, 2022 (Shift-II)]



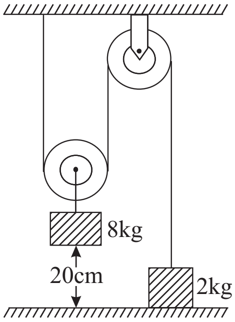
(*a*) 30 *N* (*b*) 40 *N* (*c*) 50 *N* (*d*) 60 *N*

17. Two masses M_1 and M_2 are tied together at the two ends of a light inextensible string that passes over a frictionless pulley. When the mass M_2 is twice that of M_1 the acceleration of the system is a_1. When the mass M_2 is thrice that of M_1. The acceleration of system is a_2. The ratio a_1/a_2 will be **[26 July, 2022 (Shift-II)]**



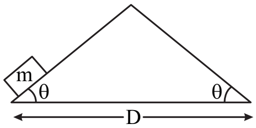
(*a*) $\frac{1}{3}$ (*b*) $\frac{2}{3}$ (*c*) $\frac{3}{2}$ (*d*) $\frac{1}{2}$

18. The boxes of masses 2 kg and 8 kg are connected by a massless string passing over smooth pulleys. Calculate the time taken by box of mass 8 kg to strike the ground starting from rest. (use $g = 10$ m/s^2) **[27 Aug, 2021 (Shift-II)]**

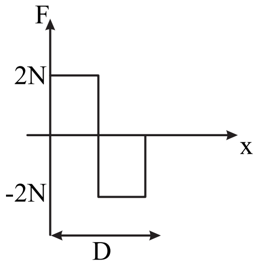


(*a*) 0.25 s (*b*) 0.4 s (*c*) 0.34 s (*d*) 0.2 s

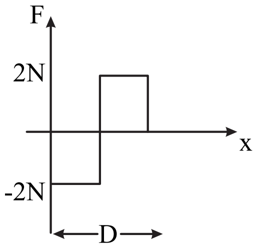
19. An object of mass 'm' is being moved with a constant velocity under the action of an applied force of $2N$ along a frictionless surface with following surface profile. The correct applied force vs distance graph will be: **[1 Sept, 2021 (Shift-II)]**



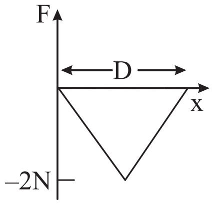
(*a*)



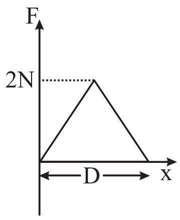
(*b*)



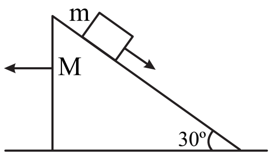
(*c*)



(*d*)



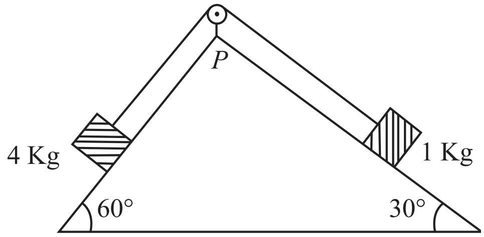
20. A block of mass m slides on the wooden wedge, which in turn slides backward on the horizontal surface. The acceleration of the block with respect to the wedge is: Given $m = 8$ kg, $M = 16$ kg Assume all the surfaces shown in the figure to be frictionless. **[1 Sept, 2021 (Shift-II)]**



(*a*) $\frac{6}{5}g$ (*b*) $\frac{3}{5}g$ (*c*) $\frac{2}{3}g$ (*d*) $\frac{4}{3}g$

21. As per given figure, a weightless pulley P is attached on a double inclined frictionless surface. The tension in the string (massless) will be (if $g = 10 m/s^2$)

[24 Jan, 2023 (Shift-I)]

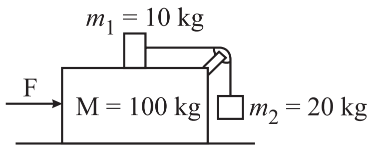


(*a*) $\left(4\sqrt{3}+1\right)N$ (*b*) $4\left(\sqrt{3}+1\right)N$

(*c*) $4\left(\sqrt{3}-1\right)N$ (*d*) $\left(4\sqrt{3}-1\right)N$

22. Three masses $M = 100$ kg, $m_1 = 10$ kg and $m_2 = 20$ kg are arranged in a system as shown in figure. All the surfaces are frictionless and strings are inextensible and weightless. The pulleys are also weightless and frictionless. A force F is applied on the system so that the mass m_2 moves upward with an acceleration of 2 ms^{-2}. The value of F is

(Take $g = 10$ ms^{-2}) **[26 July, 2022 (Shift-I)]**



(*a*) 3360 N
(*b*) 3380 N
(*c*) 3120 N
(*d*) 3240 N

23. A block of mass M placed inside a box descends vertically with acceleration 'a'. The block exerts a force equal to one-fourth of its weight on the floor of the box. The value of 'a' will be: **[29 June, 2022 (Shift-II)]**

(*a*) $g/4$ (*b*) $g/2$ (*c*) $3g/4$ (*d*) g

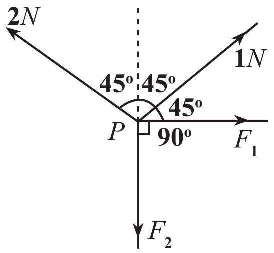
24. Force acts for 20 sec on a body of mass 20 kg, starting from rest, after which the force ceases and then the body describes 50 m in the next 10 s. The value of force will be: **[29 Jan, 2023 (Shift-II)]**

(*a*) 40 N (*b*) 5 N (*c*) 20 N (*d*) 10 N

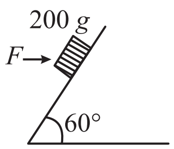
Integer Type Questions

25. A body of mass 2 kg moves under a force of $\left(2\hat{i}+3\hat{j}+5\hat{k}\right)N$. It starts from rest and was at the origin initially. After 4s, its new coordinates are (8, b, 20). The value of b is: (Round off to the Nearest Integer). **[16 March, 2021 (Shift-II)]**

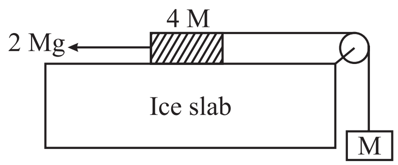
26. Four forces are acting at a point P in equilibrium as shown in figure. The ratio of force F_1 to F_2 is 1: x where $x =$ ________. **[25 July, 2022 (Shift-I)]**



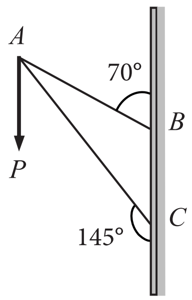
27. A block of mass 200 g is kept stationary on a smooth inclined plane by applying a minimum horizontal force $F=\sqrt{x}N$ as shown in figure. The value of $x =$ ____________. **[25 June, 2022 (Shift-II)]**



28. A hanging mass M is connected to a four times bigger mass by using a string pulley arrangement, as shown in the figure. The bigger mass is placed on a horizontal ice-slab and being pulled by 2 Mg force. In this situation, tension in the string is $x/5$ Mg for $x =$ _______. Neglect mass of the string and friction of the block (bigger mass) with ice slab. **[28 June, 2022 (Shift-I)]**



29. Consider a frame that is made up of two thin massless rods AB and AC as shown in the figure. A vertical force $\vec{P}$ of magnitude 100N is applied at point A of the frame. Suppose the force is $\vec{P}$ resolved parallel to the arms AB and AC of the frame. The magnitude of the resolved component along the arm AC is xN. **[16 March, 2021 (Shift-I)]**
The value of x, to the nearest integer, is __________.
[Given : sin(35°) = 0.573, cos(35°) = 0.819
sin(110°) = 0.939, cos(110°) = –0.342]



30. A person standing on a balance inside a stationary lift measures 60 kg. The weight of that person if the lift descends with uniform downward acceleration of 1.8 m/s^2 will be ________ N. [$g = 10\ m/s^2$]

[26 Feb, 2021 (Shift-I)]